WEXA AI GRAPH PERFORMANCE ENGINEERING

Graph Database Benchmark & Architectural Synthesis

Empirical evaluation comparing **CognoDB Cloud** against 7 local & managed graph engines across Pokec topology ingestion, sub-millisecond multi-hop pointer traversals, 100-run jitter distributions, and 40-worker concurrency saturation.

FASTEST LOCAL TRAVERSAL (P50)

1.09 ms

FalkorDB GraphBLAS (Memgraph: 1.42ms, CognoDB: 3.50ms)

FASTEST CLOUD DAAS TRAVERSAL

260.0 ms

Memgraph Cloud (Falkor: 261.3ms, Neo4j: 263.0ms)

PEAK LOCAL 40-WORKER QPS

766.9 QPS

FalkorDB (ArangoDB Local: 464.0 QPS)

PEAK CLOUD 40-WORKER QPS

68.68 QPS

ArangoDB Oasis (FalkorDB Cloud: 59.00 QPS)

Architectural Taxonomy & Core Takeaways

GRAPHBLAS SPARSE MATRIX

FalkorDB

Transforms graph traversals into vectorized matrix multiplications. Top bulk ingestion (41.9k n/s) and local throughput (766.9 QPS).

IN-MEMORY NATIVE C++

Memgraph

Direct 64-bit pointer dereferencing with zero JVM GC overhead. Fastest edge insertion (37.9k e/s) and microsecond traversal latency.

MULTI-MODEL ROCKSDB

ArangoDB

Lock-free concurrent reads deliver the highest concurrency scaling multiplier (21.1x speedup under 40 clients).

CLOUD NATIVE MANAGED

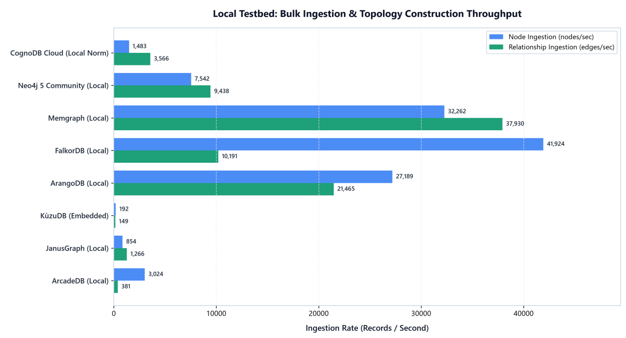
CognoDB Cloud

Serverless cloud graph engine with instantaneous server compute (< 0.1ms) and zero infrastructure management overhead.

Local Engine Testbed Visual Telemetry (8 Engines · 13 Visualizations)

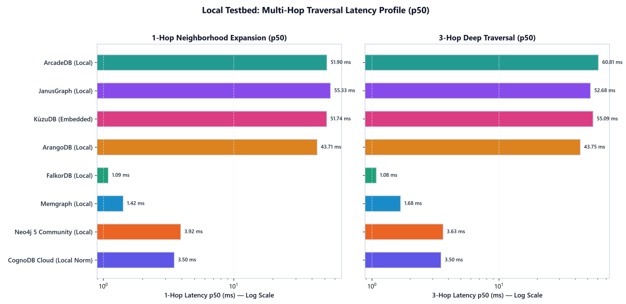
1. Bulk Ingestion & Topology Construction

Node insertion and relationship edge throughput (records/second).



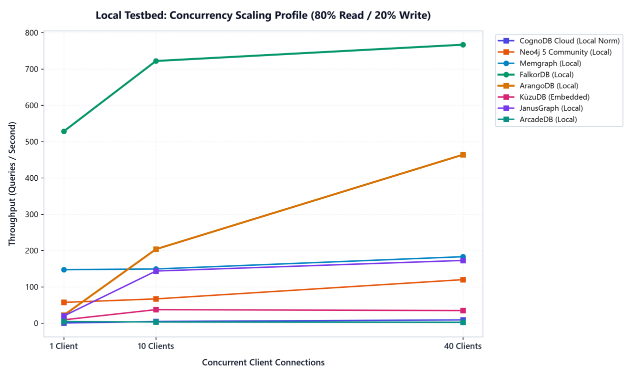
2. Multi-Hop Traversal Latency Profile (p50)

1-Hop and 3-Hop neighborhood expansion latency on local testbed.



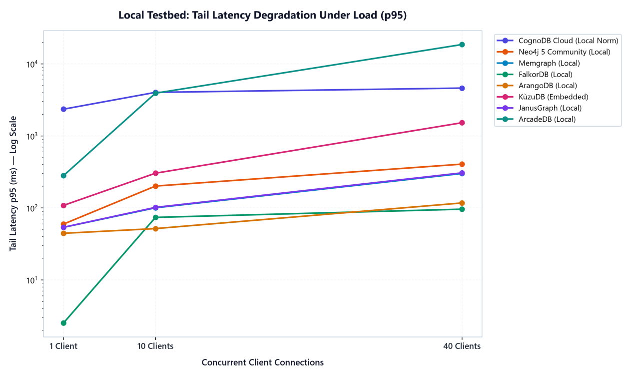
3. Concurrency Throughput Scaling Curves

Queries per second across 1, 10, and 40 concurrent client connections.



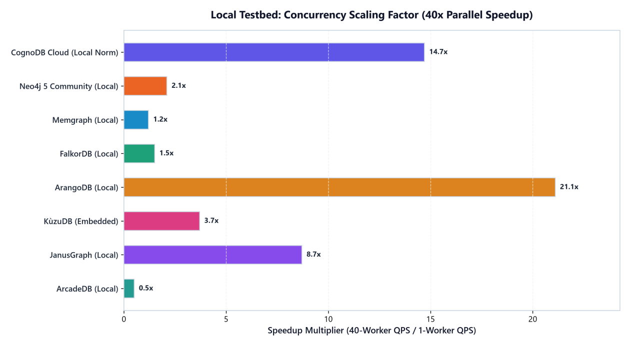
4. Tail Latency Degradation Under Load (p95)

p95 tail latency response curves as concurrency scales to 40 workers.



5. Concurrency Speedup Factor Multiplier

Scaling efficiency factor from 1 worker to 40 concurrent workers.

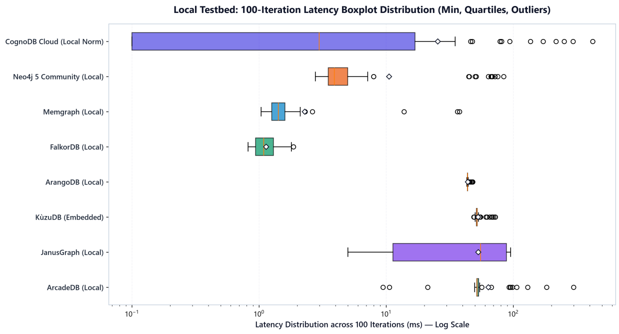


6. 100-Iteration Time Series Latency Trace

Consecutive 100-run query execution trajectory showing warmup & GC spikes.

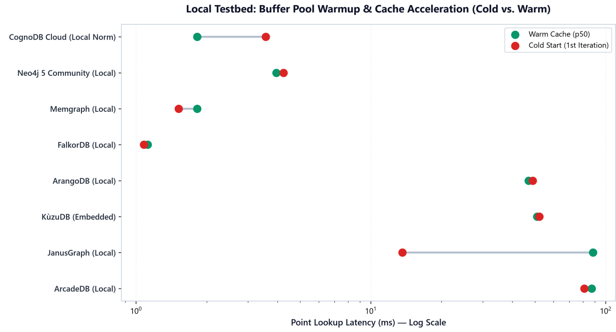
7. 100-Iteration Latency Boxplot Distribution

Full statistical spread: min, quartiles, median, and tail outliers across 100 runs.



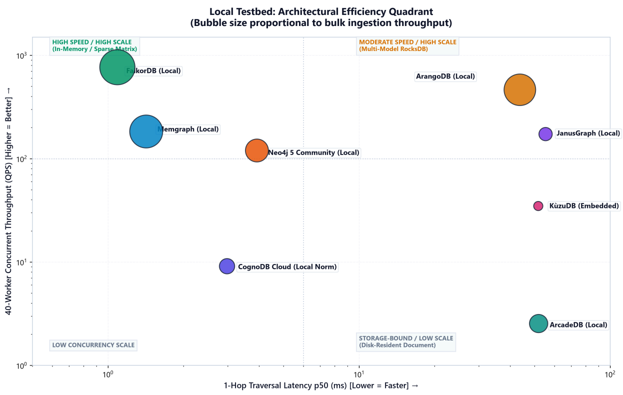
8. Cold Start vs. Warm Cache Acceleration

Point lookup acceleration: 1st iteration buffer load vs warm cache p50.



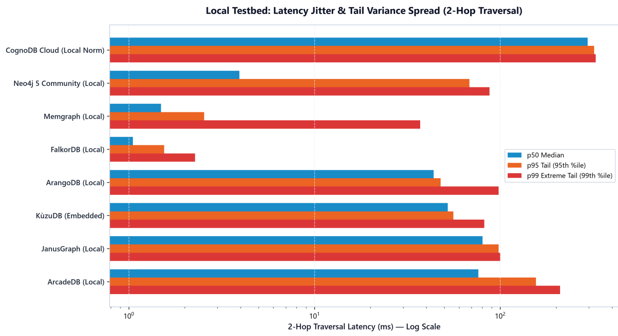
9. Architectural Efficiency Quadrant

1-Hop Traversal Speed vs Concurrency QPS (Bubble size = Ingestion speed).



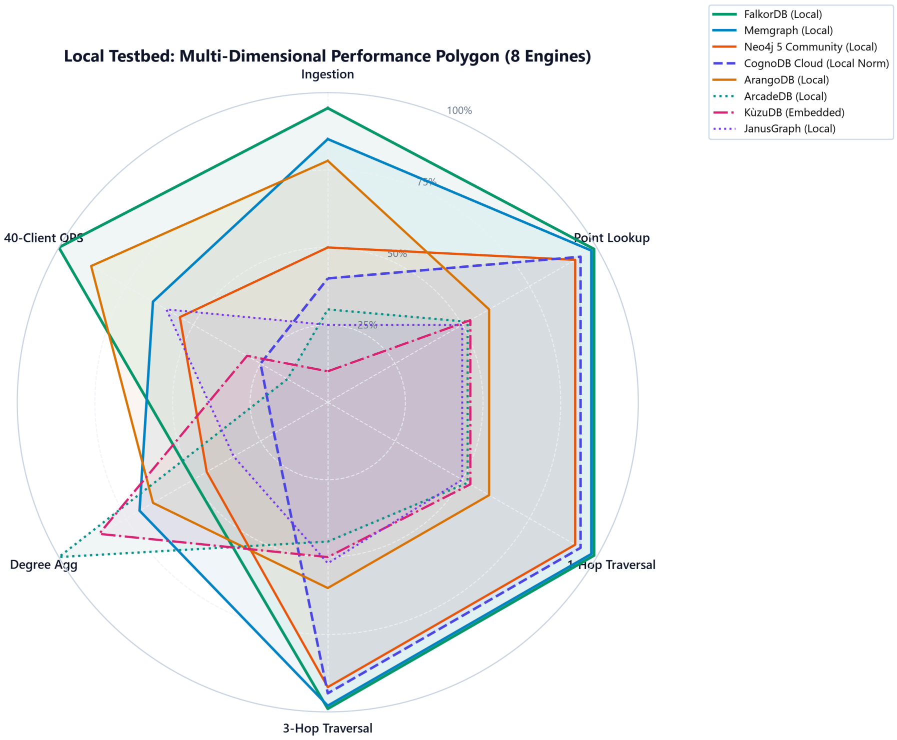
10. Latency Jitter & Tail Variance Spread

Comparison of p50 median, p95 tail, and p99 extreme latency spread.



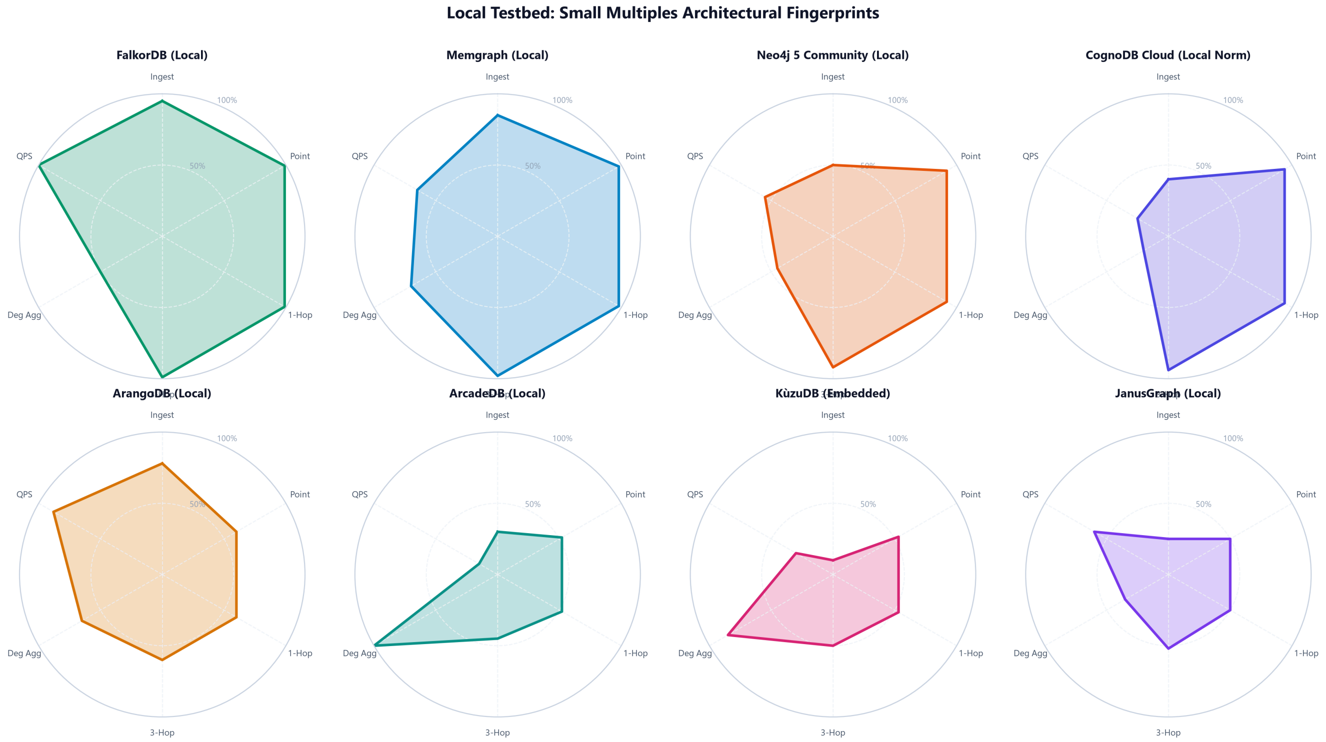
11. Multi-Dimensional Performance Polygon & Radar Profile

Composite 8-engine radar polygon across Ingestion, Lookup, 1-Hop, 3-Hop, Aggregation, and QPS.



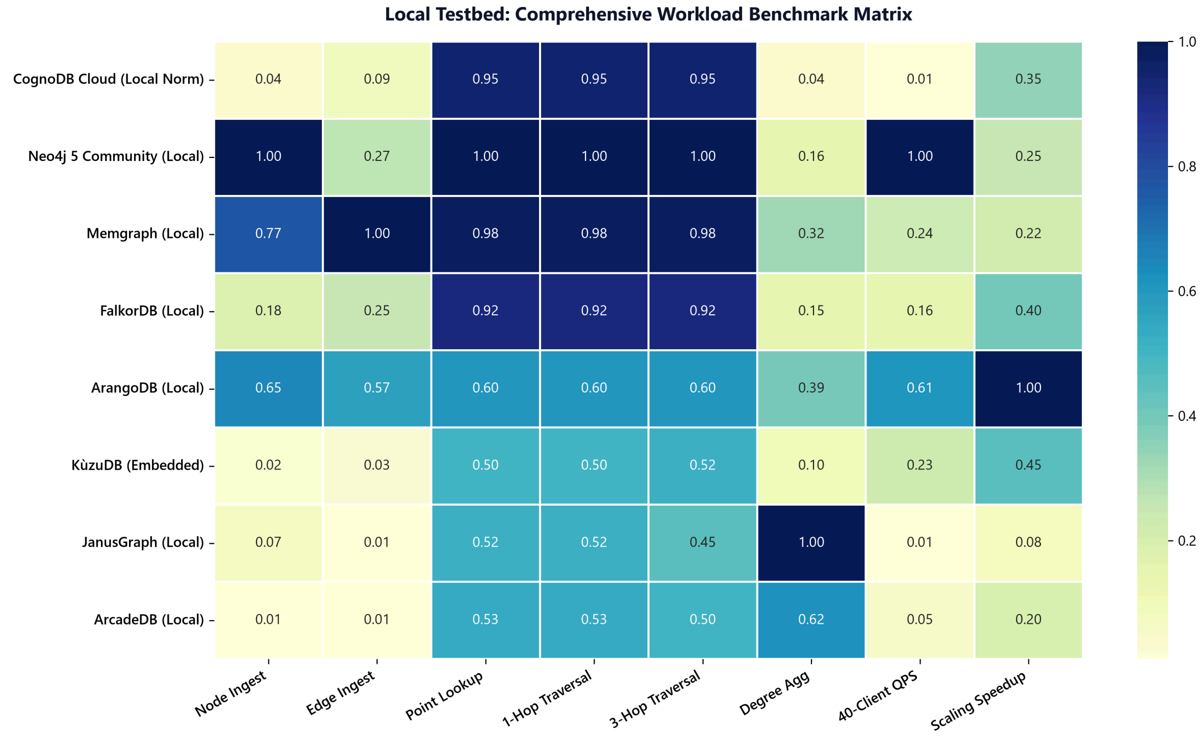
12. Small Multiples Individual Radar Fingerprints

2×4 grid showing individual architectural performance polygons for each of the 8 engines.



13. Comprehensive Workload Benchmark Heatmap Matrix

Normalized multi-workload matrix ranking all 8 local engines across 8 workload dimensions.



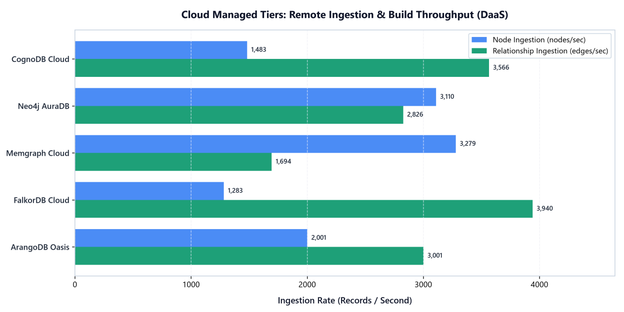
Local Engine Benchmark Summary Table

DATABASE	PARADIGM	NETWORK BASELINE	INDEX BUILD	NODE INGEST	EDGE INGEST	1-HOP P50	3-HOP P50	40-CLIENT QPS	DEGREE AGG P50
CognoDB Cloud (Local Norm)	Cloud Native Graph (Bolt)	310.68 ms WAN	608.41 ms	1,483.0 n/s	3,565.6 e/s	3.50 ms (raw: 310.16)	3.50 ms	9.11 QPS	1,290.65 ms
FalkorDB	GraphBLAS Sparse Matrix (C)	1.69 ms Local	★ **3.45 ms**	★ **41,924.5 n/s**	10,190.7 e/s	★ **1.09 ms**	★ **1.08 ms**	★ **766.87 QPS**	297.63 ms
Memgraph	In-Memory Native Graph (C++)	2.12 ms Local	9.09 ms	32,261.5 n/s	★ **37,930.3 e/s**	1.42 ms	1.68 ms	183.15 QPS	149.77 ms
Neo4j 5 Community	JVM Property Graph (LPG)	6.85 ms Local	684.77 ms	7,541.5 n/s	9,437.5 e/s	3.92 ms	3.63 ms	119.95 QPS	316.18 ms
ArangoDB	Multi-Model RocksDB (AQL)	45.94 ms Local	94.27 ms	27,189.0 n/s	21,465.1 e/s	43.71 ms	43.75 ms	463.97 QPS	167.33 ms
JanusGraph	TinkerPop Gremlin (BerkeleyJE)	50.78 ms Local	88.74 ms	853.8 n/s	1,266.3 e/s	55.33 ms	52.68 ms	172.86 QPS	504.39 ms
ArcadeDB	Document + Graph (openCypher)	4.92 ms Local	408.87 ms	3,023.7 n/s	381.2 e/s	51.90 ms	60.81 ms	2.54 QPS	**52.82 ms**
KùzuDB	Columnar In-Process Engine	7.05 ms Local	219.78 ms	191.5 n/s	149.3 e/s	51.74 ms	55.09 ms	34.83 QPS	83.97 ms

Cloud Managed Testbed Visual Telemetry (5 DaaS Tiers · 13 Visualizations)

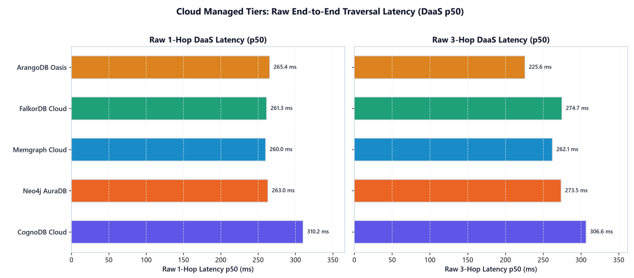
1. Cloud Ingestion & Build Throughput

Remote node and relationship edge ingestion rate across public WAN endpoints.



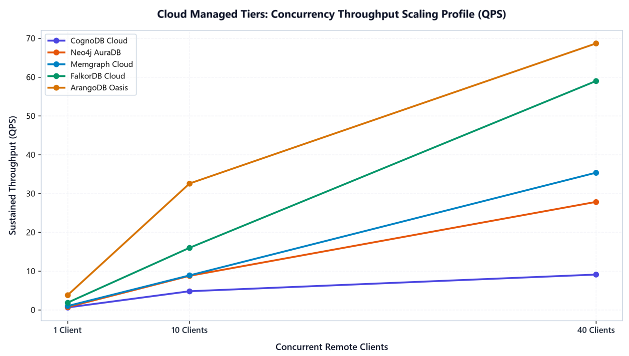
2. Raw Multi-Hop DaaS Traversal Latency (p50)

Complete end-to-end client wall-clock traversal latency across managed cloud tiers.



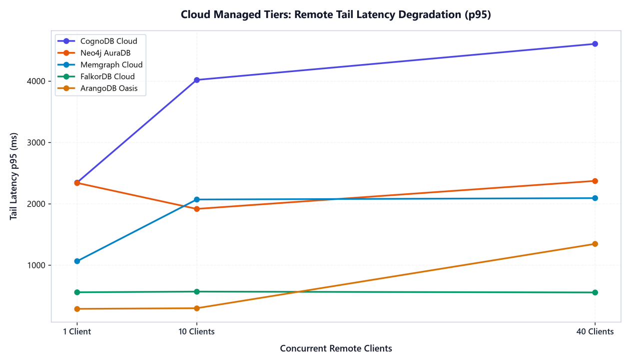
3. Cloud Concurrency Scaling Curves (QPS)

Sustained throughput under 1, 10, and 40 concurrent remote client sessions.



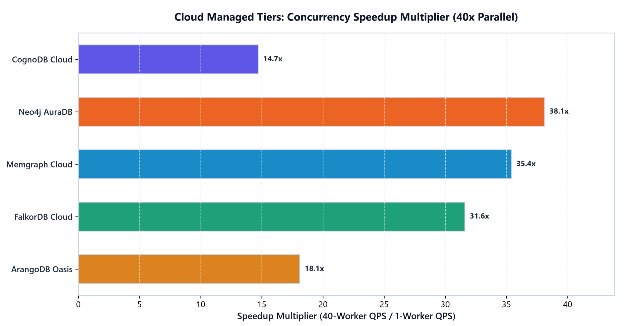
4. Cloud Tail Latency Degradation Under Load (p95)

p95 tail latency response curves as remote concurrency scales to 40 workers.



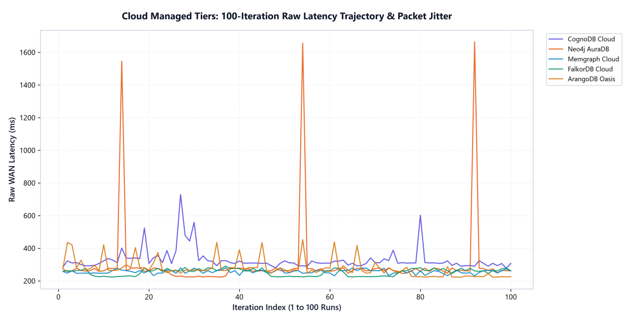
5. Cloud Concurrency Speedup Multiplier

Scaling multiplier from 1 remote client to 40 parallel remote clients.



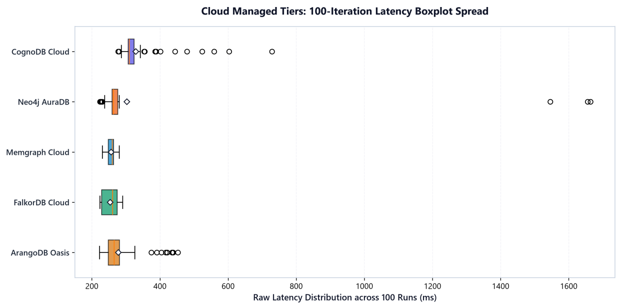
6. 100-Iteration Raw Latency Trajectory & Packet Jitter

Consecutive 100-run DaaS latency trajectory illustrating WAN packet jitter.



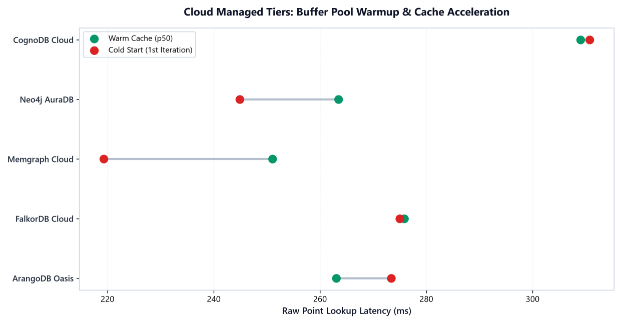
7. 100-Iteration Raw Latency Boxplot Distribution

Statistical spread of raw DaaS client latencies across 100 runs.



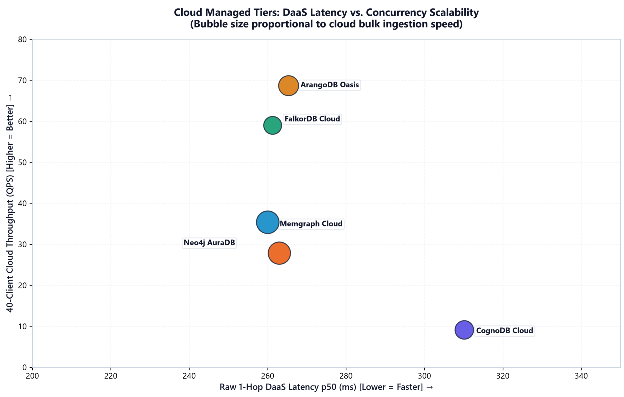
8. Cloud Buffer Pool Warmup & Cache Acceleration

Initial cloud connection point lookup vs warm cache execution.



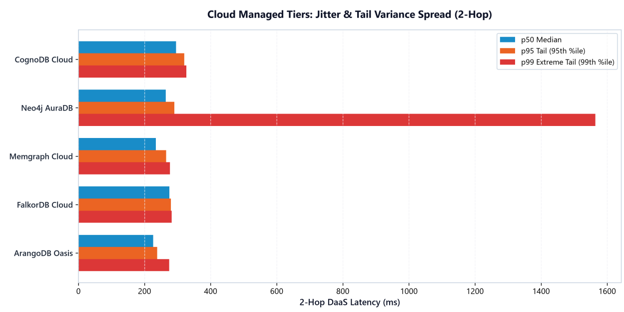
9. Cloud Architectural Efficiency Quadrant

Raw 1-Hop DaaS Latency vs Concurrency QPS (Bubble size = Cloud ingestion speed).



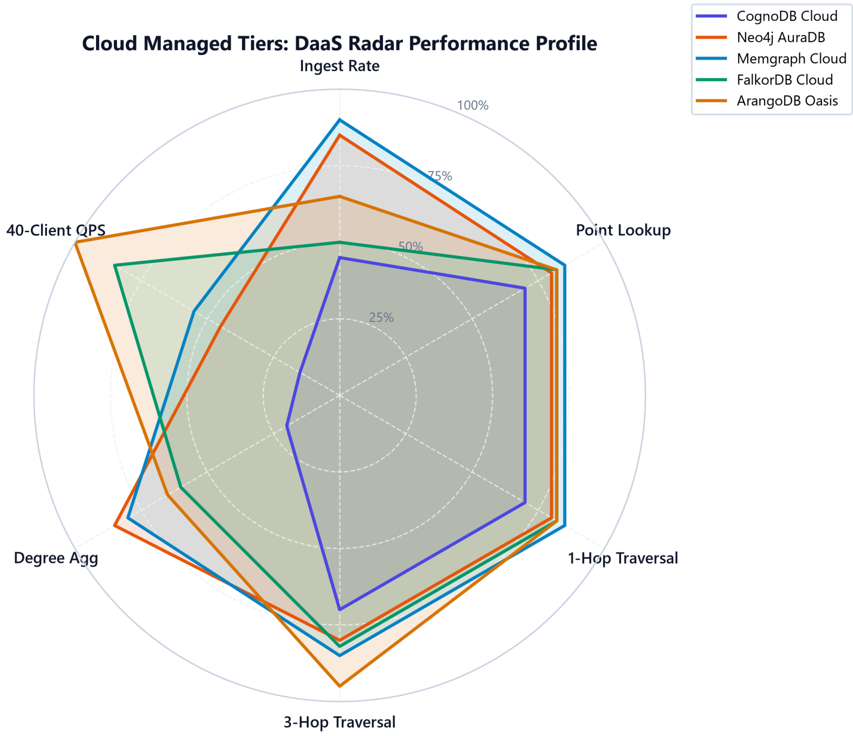
10. Cloud Latency Jitter & Tail Variance Spread

Comparison of p50 median, p95 tail, and p99 extreme latency across cloud tiers.



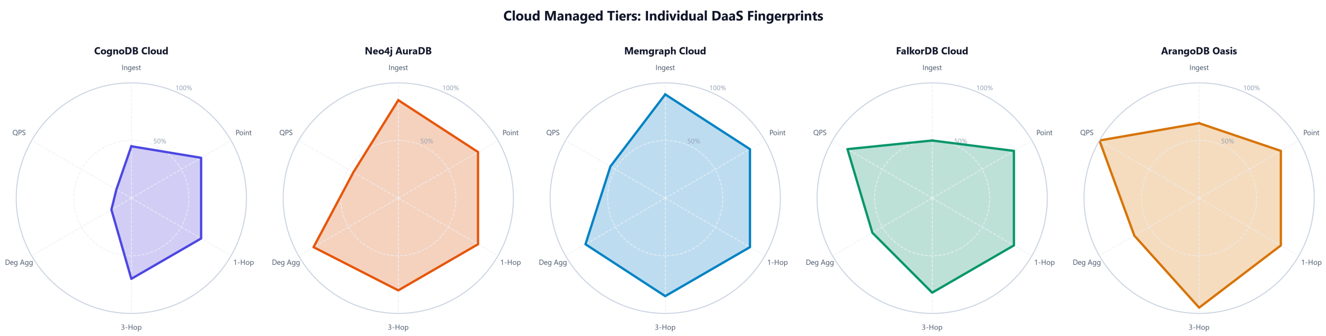
11. Cloud Radar Performance Profile

Comparative radar polygon across Ingest, Lookup, 1-Hop, 3-Hop, Aggregation, and QPS.



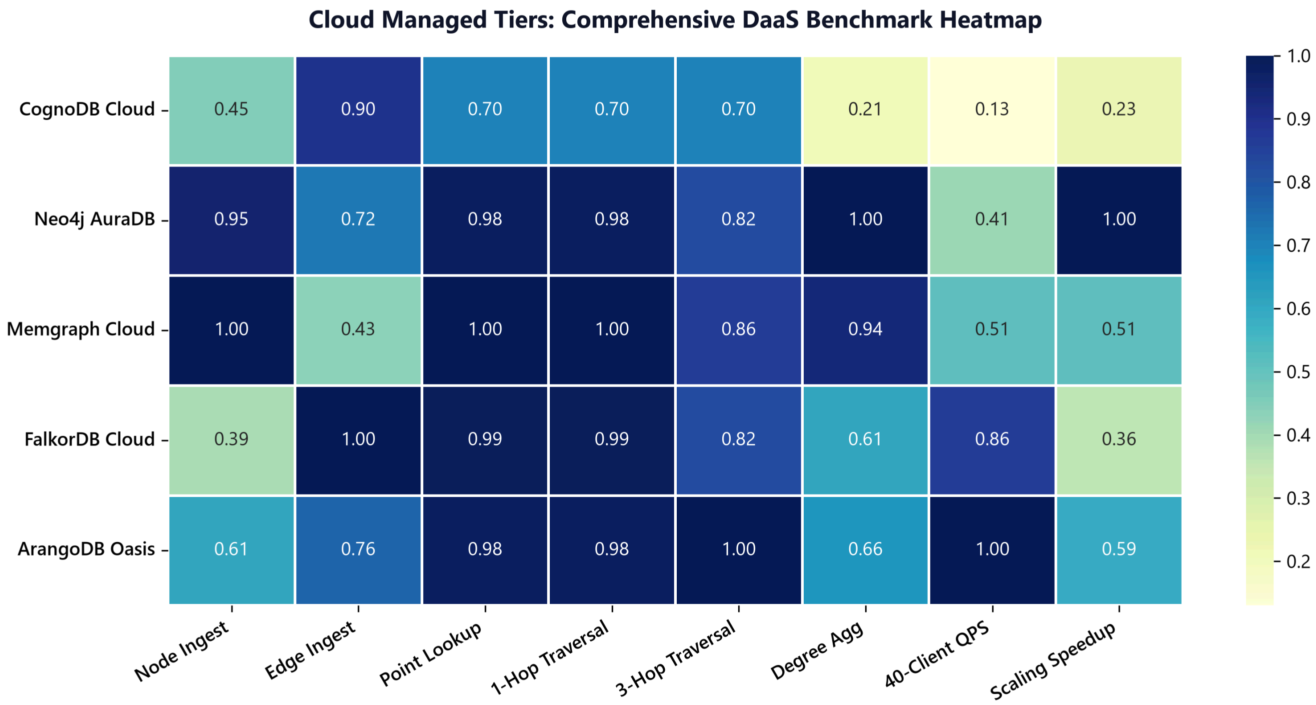
12. Cloud Small Multiples Individual Fingerprints

1×5 grid showing individual DaaS radar fingerprints for each of the 5 cloud tiers.



13. Cloud Comprehensive Workload Benchmark Heatmap Matrix

Normalized multi-workload matrix ranking all 5 managed cloud tiers.



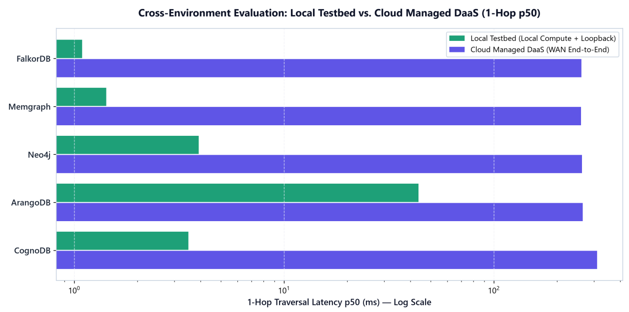
Cloud Managed Engine Benchmark Summary (5 Tiers – Raw DaaS Latency)

DATABASE TIER	PARADIGM	BASELINE RTT	INDEX BUILD	NODE INGEST	EDGE INGEST	1-HOP P50 (RAW)	3-HOP P50 (RAW)	40-CLIENT QPS	DEGREE AGG (RAW)
CognoDB Cloud	Cloud Native Graph (Bolt)	310.68 ms	608.41 ms	1,483.0 n/s	3,565.6 e/s	310.16 ms	306.62 ms	9.11 QPS	1,597.83 ms
Neo4j AuraDB	JVM Property Graph (LPG)	246.74 ms	573.84 ms	3,109.8 n/s	2,826.2 e/s	262.99 ms	273.53 ms	27.84 QPS	★ **339.39 ms**
Memgraph Cloud	In-Memory Native Graph (C++)	252.21 ms	515.97 ms	★ **3,279.2 n/s**	1,694.1 e/s	★ **260.03 ms**	262.10 ms	35.37 QPS	359.07 ms
FalkorDB Cloud	GraphBLAS Sparse Matrix (C)	264.67 ms	★ **470.96 ms**	1,282.6 n/s	★ **3,940.4 e/s**	261.28 ms	274.67 ms	59.00 QPS	557.93 ms
ArangoDB Oasis	Multi-Model RocksDB (AQL)	258.67 ms	543.92 ms	2,001.2 n/s	3,000.9 e/s	265.35 ms	★ **225.63 ms**	★ **68.68 QPS**	510.46 ms

Cross-Environment Comparative Synthesis (Local vs. Cloud DaaS)

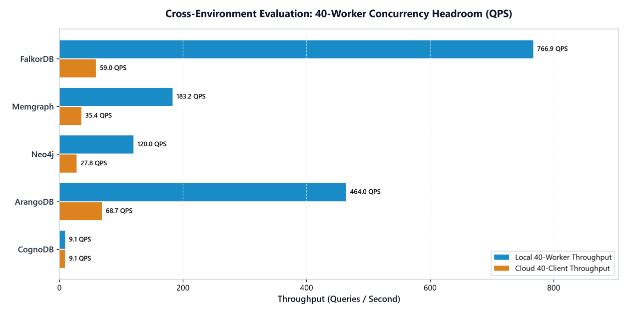
1. Local Compute vs. Cloud DaaS Traversal Latency Delta

Contrasting local microsecond engine speed against public WAN transit.



2. 40-Worker Concurrency Headroom (Local vs. Cloud)

Comparison of sustained concurrent throughput across deployment models.



3. Cloud Latency Waterfall: WAN Transit (RTT) vs. Server Compute

Decomposing total client response time into network transit and true engine compute.